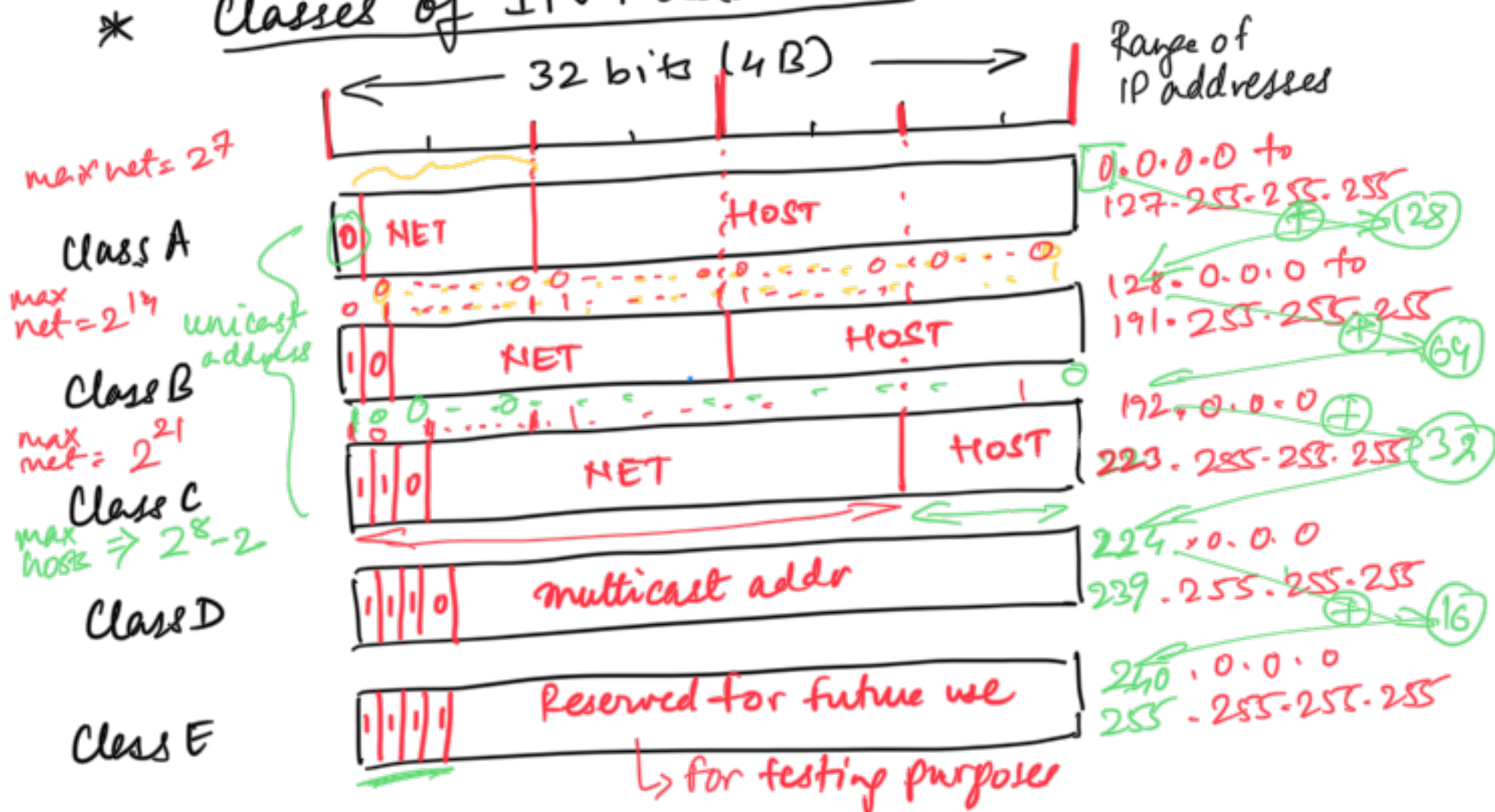


RECAP

IPv4: addr size = 32 bits

* Classes of IPv4 addresses



* PRIVATE IP ADDRESS RANGE

Class A - 10.0.0.0 to 10.255.255.255

Class B - 172.16.0.0 to 172.31.255.255

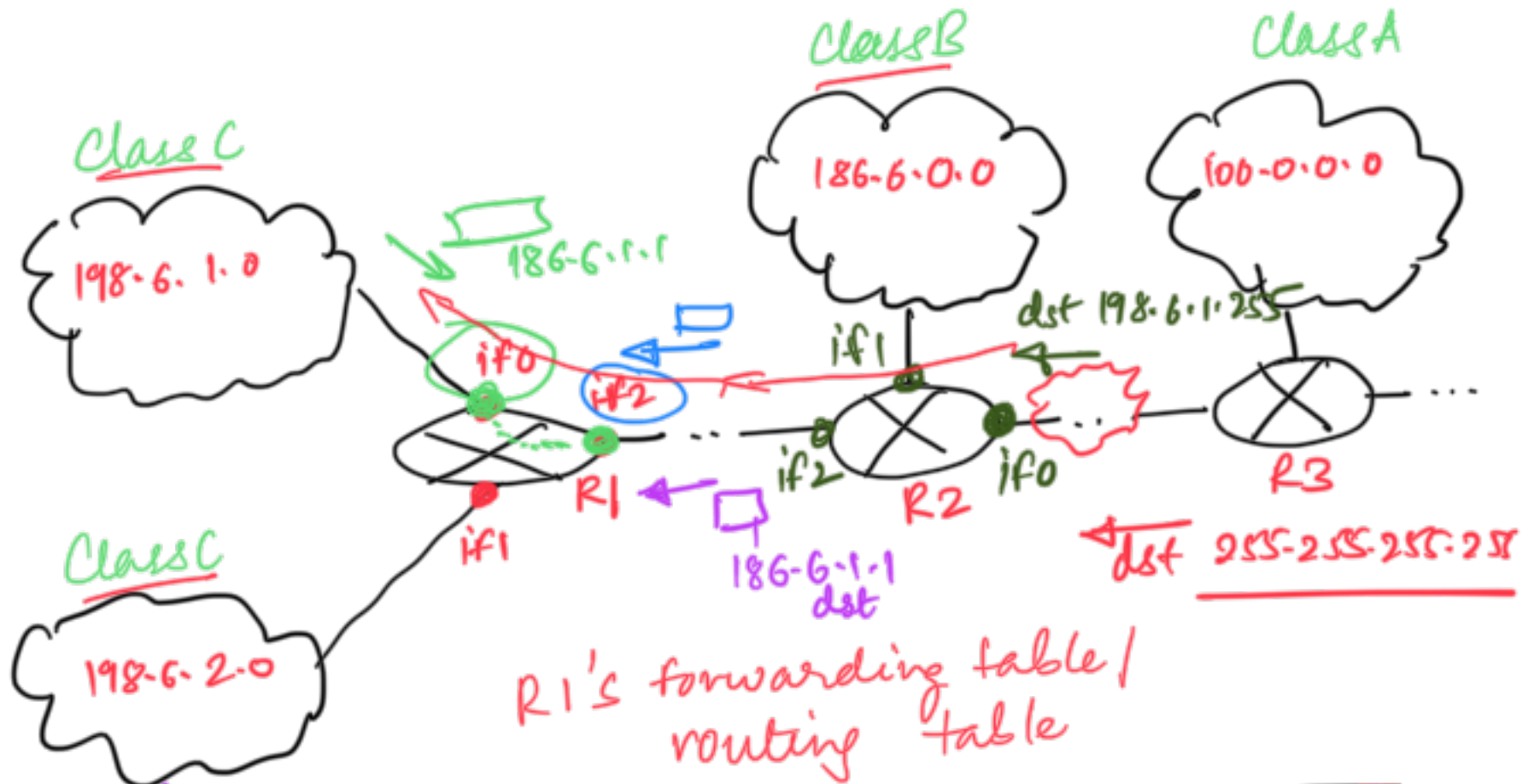
Class C - 192.168.0.0 to 192.168.255.255

* Max no. of networks = Fill this

* Max no. of hosts on a network = Fill this

	NET ID	HOST ID
Network address	SPECIFIED	ALL 0's
Broadcast address	SPECIFIED ALL 1's	ALL 1's
Subnet mask	ALL 1's	ALL 0's

Q. How does a router obtain network address for an incoming packet?



R1's forwarding table / routing table

① parse IP header

② Translate IP to net addr

IP addr
 $\wedge$ Subnet mask
 Net addr

186.6.1.1
 $\wedge$ 255.255.255.0
 186.6.1.0

186.6.1.1
 $\wedge$ 255.255.0.0
 186.6.0.0

Network addr	Subnet mask	Next hop	Cost...
198.6.1.0 24	255.255.255.0	if0	..
198.6.2.0 24	255.255.255.0	if1	..
186.6.0.0 16	255.255.0.0	if2	..
100.0.0.0 8	255.0.0.0	if2	..

* Special IP address.

① Network address
Netid
XX

Hostid
All 0's

② Broadcast address

→ Local broadcast
All 1's

All 1's

→ Broadcast on a specific network
Specified All 1's

③ This host $\Rightarrow$ 0.0.0.0 $\Rightarrow$ during boot up
Netid Hostid
All 0's All 0's

④ Loopback IP address --- for testing
127.XX.YY.ZZ
localhost: 127.0.0.1